INCH POUND

MIL-C-44413
9 August 1990
(See 6.9)

MILITARY SPECIFICATION

COVER, NUCLEAR, BIOLOGICAL, AND CHEMICAL PROTECTIVE (NBC-CP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a lightweight disposable (low cost) cover to protect exposed supplies and equipment from nuclear/chemical/biological attack and serves to ease the burden of decontamination throughout the Army.

1.2 Classification. The cover shall be of the following types and sizes as specified (see 6.2):

Type I - Camouflage Green 483
Type II - Desert Tan 459
Type III - Reversible Camouflage Green 483/White

Size 1 - 12 feet wide by 309 feet long
Size 2 - 20 feet wide by 149 feet long

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

CRDC-SP-84010 - Laboratory Methods for Evaluating Protective Clothing Systems Against Chemical Agents

(Copies are available from the Chemical Research and Development Center, ATTN: DRSMC-CLW-P(A), Aberdeen Proving Ground, MD 21010.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 904 - Comparison of Bond Strength or Ply Adhesion of Similar Laminates Made from Flexible Materials
D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus
D 2103 - Polyethylene Film and Sheeting

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The cover shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Base cloth. The base cloth shall be a plain woven fabric of yarn specified in 3.3.2. The yarn shall be either 8 by 8 per inch or 10 by 10 per inch in the warp and filling directions. The weight of the base cloth shall be 3.0 $\pm$ 0.5 ounces per square yard.

3.3.2 Yarn. The yarn for the warp and filling of the fabric shall be slit film ribbons of high density polyethylene polymer, 900 $\pm$ 50 denier, suitably compounded, oriented, and heat stabilized to meet the requirements of the laminated cloth (see table I).

3.3.3 Coating compound. The coating compound shall be a low density polyethylene coating, suitably compounded and pigmented such that the finished, coated and laminated cloth conforms to the requirements of table I.

3.3.4 Cloth, coated. The base cloth specified in 3.3.1 shall be coated on each side with 1.5 $\pm$ 0.25 mil thickness of coating compound specified in 3.3.3.

3.3.5 Film, laminated. The laminated film shall be 2 $\pm$ 0.25 mils thick (see 6.10), and be suitably compounded such that the finished coated and laminated cloth conforms to the requirements of table I.

3.4 Cloth, coated and laminated. The coated and laminated cloth shall be constructed so that it is layered in the following progression: The coated cloth, as specified in 3.3.4, shall be laminated to the laminated film

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specified in 3.3.5. The laminated film shall then be coated with the coating compound specified in 3.3.3, at a thickness of 1.5 ± 0.25 mils. The face of the finished coated and laminated cloth shall be the side with the laminated film. The coated and laminated cloth shall conform to the requirements specified in table I, when tested as specified in 4.4.4.

TABLE I. Physical requirements

Characteristics	Requirements Min.	Max.
Cloth, coated, laminated:		
Weight, ounces per square yard	7.0	10.0
Thickness, mils	7.5	10.5
Breaking strength, pounds:		
Warp	135	--
Filling	135	--
Tearing strength, grams:		
Warp	3000	--
Filling	3000	--
Hydrostatic resistance, psi (face to water)	175	--
Resistance to abrasion, percent strength loss:		
Warp	--	20
Filling	--	20
Ply, adhesion, pounds per inch	7	--
Flammability:		
After flame time, seconds:		
Warp	--	6
Filling	--	6
After glow time, seconds:		
Warp	--	0
Filling	--	0
Resistance to toxicological agents, breakthrough time, hours:		
Mustard (HD)	72	168
Thickened GD	72	168

3.5 Color. The color of the cover shall be Camouflage Green 483 (matching color chip 34094 of FED-STD-595), Desert Tan 459 or a reversible Camouflage Green 483/White, as specified.

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3.5.1 Matching. The color of the finished cover shall match the standard sample (see 6.4) when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles. The color shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.6 Spectral reflectance.

3.6.1 Camouflage Green 483. When Camouflage Green 483 is specified, the spectral reflectance values for visible/near infrared wavelengths of the finished cloth shall meet the values specified in table II, when tested as specified in 4.4.4

TABLE II. Spectral reflectance (percent) for Camouflage Green 483

Wavelength nanometers	Reflectance Max.	Min.	Wavelength nanometers	Reflectance Max.	Min.
600	12	5	740	42	30
620	12	5	760	50	39
640	12	5	780	56	45
660	13	5	800	60	47
680	15	7	820	63	49
700	21	12	840	65	50
720	30	20	860	66	51

3.6.2 Desert Tan 459. When Desert Tan 459 is specified, the spectral reflectance values for visible/near infrared wavelengths of the finished cloth shall meet the values specified in table III, when tested as specified in 4.4.4.

TABLE III. Spectral reflectance (percent) for Desert Tan 459

Wavelength nanometers	Reflectance Max.	Min.
700	65	45
720	65	45
740	65	45
760	65	45
780	65	45
800	65	45
820	65	45
840	65	45
860	65	45

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3.7 Width.

3.7.1 Size 1. The overall width of the cover, after selvage trimming, shall be not less than 12 feet (see 3.12). The cover shall be constructed to form the specified width using a maximum of three seams.

3.7.2 Size 2. The overall width of the cover, after selvage trimming, shall be not less than 20 feet (see 3.12). The cover shall be constructed to form the specified width using a maximum of five seams.

3.8 Seams. All seams shall be extrusion sealed. The cloth shall overlap 1 ± 1/4 inch. The extrusion seal shall be 1 ± 1/4 inch in width. Material contained in the extrusion seal shall be 1 to 2 mils in thickness.

3.9 Splicing. No splicing shall be allowed in the lengthwise direction on either size of cover in not more than 10 percent of an order. No roll shall contain more than one splice. The splice length shall be not less than 20 feet.

3.10 Marking. All edges in the length direction shall be marked with a brightly colored, removable tape every 39.3 inches. The tape shall be at least 6 ± 1/2 inches long and 3/4 inch wide. The tape shall be placed perpendicular to the lengthwise edge of the cover.

3.11 Warning tag. A warning tag shall be attached to the edge of the roll with not finer than 5-ply cotton string doubled to not less than 8 ± 1/2 inches long. The warning tag shall be made of not less than 15 point paper stock and the finished surface shall not be calendered or pigmented to a degree which will cause the obliteration of printed, stamped, or typed markings. The warning tag shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water insoluble ink with the information shown in figure 1. The height of the lettering shall be 1/4 ± 1/16 inch high. All the information on the warning tag shall be printed, stamped, or typed. Hand lettered information is permitted, but hand written information is not permitted.

3.12 Length. For government procurements only, the cover shall be put up in full width rolls as specified in 5.1.1 and shall be furnished in lengths and widths as specified below.

Size 1 - 309 feet ± 36 inches in length and 12 feet ± 6 inches in width
Size 2 - 149 feet ± 18 inches in length and 20 feet ± 6 inches in width

3.13 Workmanship. The finished cover shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3, and tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase specification.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed below conform to the specified requirements.

Characteristics	Requirement paragraph	Test method
Plain weave	3.3.1	Visual
Construction	3.3.1	5050 1/
Weight	3.3.1	5041 1/
Yarn identification	3.3.2	--
Yarn denier	3.3.2	--

1/ Specified methods shall be in accordance with FED-STD-191.

4.4.2 End item visual examination. The completed covers shall be examined for the defects listed in table IV. Each cover in the sample unit shall be examined on one side only, however, the other side shall be alternated so that the alternate covers are examined on the face and back, respectively. The defects shall be counted regardless of their proximity to each other, except where two or more represent a single local condition of the cover, in which case only the most serious defect shall be counted. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Materials:			
General	Any cover not fabricated of specified material (unless otherwise indicated herein)	101	
Cloth	Cut, hole or tear, which extends through or exposes the base cloth	102	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Cloth (cont'd)	Abrasion mark or scratch which extends through the coating to a point where base cloth yarns or fabric weave can be seen	103	
	Any pinhole	104	
	Any uncoated area	105	
	Pit, blister, tunnel, or delamination of coating	106	
	Lump or imbedded foreign matter which protrudes so as to be readily abraded from material or easily removed	107	
	Crease or wrinkle resulting in adhesion of surface or delamination of coating when corrected by manual pressure	108	
	Coating or laminate improperly applied (i.e. not visibly heavier on one side)	109	
	Laminated construction improperly applied (i.e., not visibly heavier on one side)	110	
	Any tackiness of soft coating (coating will adhere and not unroll readily)	111	
	Ripples, waviness or dimensional distortion		201
	Edges not straight and uniform or edges folded, rolled or scalloped		202
	Spliced cloth (base cloth or coated) in more than 10 percent of order	112	
	More than one splice per roll	113	
	Splice length less than 20 feet	114	
	Impression in coating from splicing threads (except as normally occurs in manufacturing)	115	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams	Any seam (when specified) not sealed as specified	116	
	Seam overlapped less than 3/4 inch	117	
	Seam overlapped more than 1-1/2 inches		203
	Any open seam	118	
Marking	Edges not marked	119	
	Edges not marked as specified		204
	Tape not as specified		205
Assembly, general	More than three seams in construction of size 1 cover		206
	More than five seams in construction of size 2 cover		207
Shade	Shade not comparable to standard or approved sample		208
Cleanliness	Objectionable odor 1/		209
	Overall uncleanliness		210
	Any spot, stain, or streak more than 1 inch long which cannot be readily removed with naptha or water 2/	120	

1/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

2/ Clearly visible at normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The cover shall be examined for the measurement defects specified below. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Defects

Length not within specified tolerances.

Width not within specified tolerances.

Length between marking tapes not as specified.

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4.4.4 End item testing. The end items shall be tested for the characteristics listed in table V. The physical values specified in section 3 apply to the average of the determination(s) made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of covers. The sample unit shall be one finished cover. The lot shall be unacceptable if one or more sample unit fails to meet any test requirement specified. All test reports shall contain the individual values used in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

Lot size (covers)	Sample size (sample units)
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method
Cloth, coated/ laminated:		
Weight	Table I	5041 1/
Thickness	Table I	ASTM D 2103
Breaking strength	Table I	5100 1/
Tearing strength	Table I	ASTM D 1424
Hydrostatic resistance (face of fabric to water)	Table I	5512
Resistance to abrasion	Table I	5304 1/ 2/
Ply, adhesion	Table I	ASTM F 904
Flammability:		
after flame time (sec)	Table I	5903 1/
after glow time (sec)	Table I	5903 1/
Resistance to toxicological agents, hours:		
Mustard (HD)	Table I	4.5.1.1 3/
Thickened (GD)	Table I	4.5.1.2 3/
Spectral reflectance	3.6	4.5.2

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1/ Specified methods shall be in accordance with FED-STD-191.

2/ Except as follows:

a. Five specimens from each of the warp and filling directions shall be tested.

b. The face side of the finished, coated and laminated cloth shall be abraded.

c. The specimens shall be abraded for 300 continuous cycles under a tension of 5 pounds and a load of 3 pounds. The abradant shall be 3M Company, 240 grit aluminum oxide abrasive cloth or equal (see 6.8). Percent loss of strength shall be determined by Method 5102 of FED-STD-191 using a 1 inch wide specimen.

3/ Specified methods are detailed in Chemical Research Development Center Test Report CRDC-SP-84010.

4.5 Methods of inspection.

4.5.1 Toxicological agent resistance testing. Government acceptance testing shall be performed for toxicological agents resistance (see 6.5). The sample size for testing shall be 22 random samples per lot. Each sample shall consist of a 12 by 12 square inch piece of material. One sample unit should be taken from each cover sampled, unless there are 22 or fewer covers in a lot. Testing shall be in accordance with CRDC-SP-84010 and as specified in 4.5.1.1 and 4.5.1.2.

4.5.1.1 Resistance to mustard (HD). Resistance to permeation by mustard (HD) shall be determined as specified in CRDC-SP-84010 procedure 2.3 liquid contamination/liquid penetration test (Mandrel Test), except that the chemical agent challenge shall be 10 grams per square meter applied in droplets on the film surface. Test specimens shall be obtained by cutting two, 2 by 6 square inch specimens from each of the 22 samples with one determination made on the back side of each specimen. No test specimen shall be taken within 2 inches of the selvage edge. A weight (60 grams) will be attached to each end of each sample and the samples will be draped over the M8 detector paper on the mandrels. All specimens shall meet the mustard toxicological requirement listed in table I. In the event that one of the two tested specimens meet the requirements but one does not, two additional test specimens from the samples shall be tested. If both specimens meet the requirements, the material shall be considered acceptable.

4.5.1.2 Resistance to thickened GD. Resistance to permeation by GD shall be determined as specified in CRDC-SP-84010 procedure 2.3 liquid contamination/liquid penetration test (Mandrel Test), except that the chemical agent challenge shall be 10 grams per square meter applied in droplets. Test specimens shall be obtained by cutting two, 2 by 6 square inch specimens from

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each of the 22 samples with one determination made on the back side of each specimen. No test specimen shall be taken within 2 inches of the selvage edge. A weight (60 grams) will be attached to each end of each sample and the samples will be draped over the M8 detector paper on the mandrels. All specimens shall meet the GD toxicological requirements listed in table I. In the event that one of the two tested specimens meet the requirements but one does not, two additional test specimens from the samples shall be tested. If both specimens meet the requirements, the material shall be considered acceptable.

4.5.2 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for Camouflage Green 483 and from 700 to 860 nanometers (nm) for Desert Tan 459 at 20 nm intervals on a spectrophotometer (see 6.6) relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white; e.g., Halon, magnesium oxide or vitrolite titles (see 6.7). The spectral bandwidth shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CID Source A or CIE Source D65. The specimen shall be measured as a single layer backed with two layers of the same fabric and shade. Measurements shall be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. When the measured reflectance values at four or more wavelengths do not meet the limits specified in 3.6.1 or 3.6.2, it shall be a test failure.

4.5.3 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

5. PACKAGING

5.1 Put-up and preservation. Each protective cover shall be rolled to a width of approximately 5 feet +6, -0 inches, and in such a manner that the length markers are visible. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A and commercial. Each protective cover shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A, B and commercial. Each protective cover shall be in accordance with the applicable requirements of PPP-P-1136.

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5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cover is intended for use as a barrier to protect exposed supplies and equipment from chemical and biological attack and serve to ease the burden of decontamination throughout the army.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Type and size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Toxicological tests. Because of the extreme dangers involved in handling toxicological warfare agents, and the special equipment required in connection with their use, all document tests involving toxicological agents will be performed by the U.S. Army Armament Munitions and Chemical Command (Edgewood Area), Chemical Testing Branch, Building E5100, Aberdeen Proving Ground, MD 21010-5423 or by an authorized independent Testing Laboratory approved by the U.S. Army Armament Munitions and Chemical Command.

6.6 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR, and MacBeth 1500 with IR option.

6.7 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same

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source has available magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Bureau of Standards of the instrument manufacturers.

6.8 Equal item. Prior to the use of an "or equal item", the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.9 Supersession data. This document includes the requirements of Limited Production Purchase Description 2-88A, dated 6 June 1989, Cover, Nuclear, Biological, and Chemical Protective (NBC-PC).

6.10 Laminated film. The requirements for the laminated film have been met with the use of Dow Chemical's Saranex R[R], type 14.

6.11 Subject term (key word) listing.

Toxicological agents protection

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8415-0696)

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activities:

Air Force - 45
Navy - SH

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EXAMPLE IS APPROXIMATE TAG SIZE DESIRED

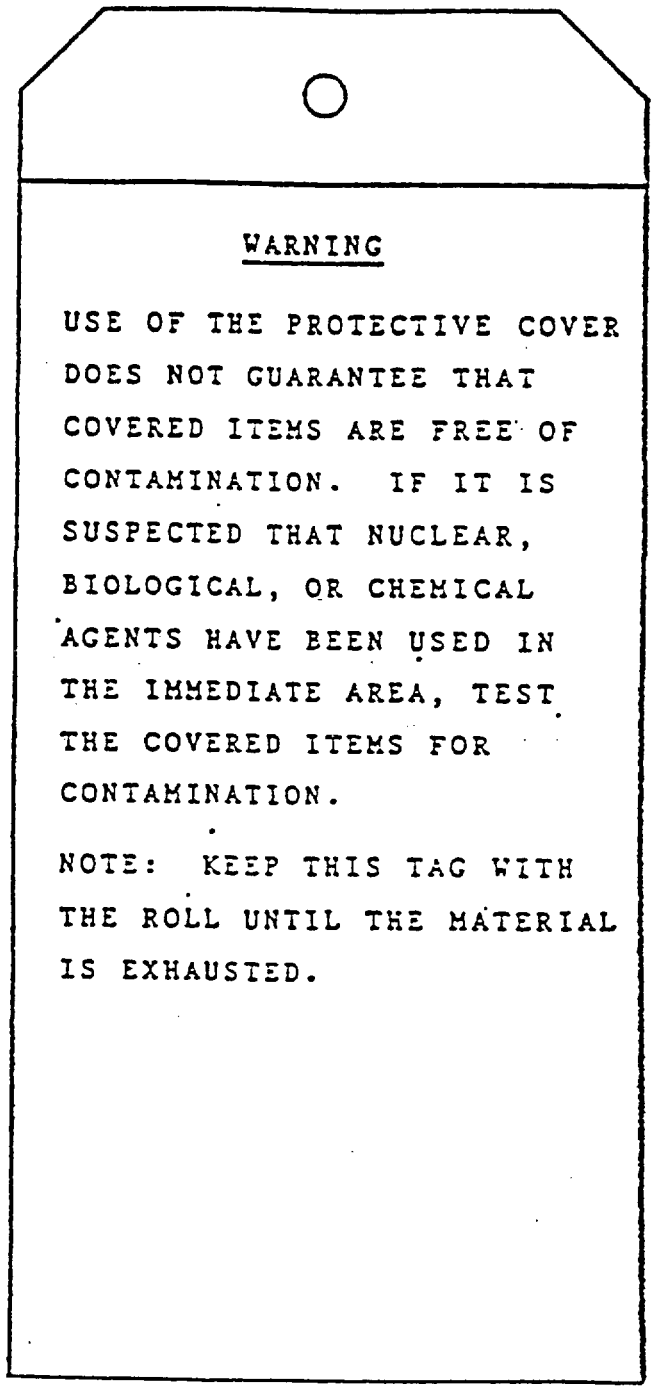


FIGURE 1. WARNING TAG

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